nature climate change

Article

https://doi.org/10.1038/s41558-025-02352-8

Using deep learning to generate key variables in global mitigation scenarios

Received: 5 August 2024

Accepted: 28 April 2025

Published online: 13 June 2025

Check for updates

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Integrated assessment models (IAMs) are the dominant tools for projecting mitigation scenarios. However, IAM-based scenarios often face challenges such as modelling biases and large computational burden. Here we develop a deep learning framework to generate key variables through synthetic mitigation scenarios aligned with the Sixth Assessment Report (AR6) Scenarios Database. By analysing 1,202 scenarios from a diverse set of IAMs, we select key drivers that enable a more detailed sectoral representation. Next, we trained three generative deep learning models to produce 30,000 synthetic scenarios at low computational cost across various IPCC AR6 climate categories, replicating variable distributions and correlations while also demonstrating physical consistency in power sector variables through internal validation checks. We found that the variational autoencoder achieved the highest label transferring accuracy among three frameworks. This study illustrates the potential of deep learning to complement IAM approaches and provides a basis for handling complex mitigation scenario generation tasks.

Climate change poses a major threat to human society, leading to environmental damage and economic losses[1,2]. Understanding the interactions between humans and the Earth's systems[3], particularly how human activities contribute to GHG emissions, is crucial in developing effective climate change mitigation strategies for stabilizing climate change. Integrated assessment models (IAMs) are central to this effort, projecting a range of alternative futures to explore the climate implications of different developmental paths and the outcomes of societal actions. These models integrate socioeconomic[4], energy[5–7], land[8], water[9] and climate systems[10], offering complex interactions that underlie the future temperature projections. Such an approach is critical for understanding the broad impacts of human actions on climate and informing future policy choices[11,12].

Over the past few decades, a wide range of climate change mitigation scenarios has been developed through IAMs[13–16] and this diverse set of scenarios has been critical for supporting robust decision-making and understanding uncertainties[17–19]. A scenario is a quantified representation of a potential future, describing how socioeconomic and technological drivers influence GHG emissions and climate outcomes under specific assumptions[20]. Accordingly, the outputs from these models are instrumental in facilitating multi-model comparisons[21–23], model diagnostics[24] and the creation of large-scale scenario ensembles[25,26]. A prime example of this is the development of the large-scale Sixth Assessment Report (AR6) Scenarios Database supporting the Intergovernmental Panel on Climate Change's Sixth Assessment Report[27], which has been crucial in understanding mitigation scenarios for international climate policymaking[28,29]. A detailed review of mitigation scenarios is provided in Supplementary Text 1.

However, assessing ensembles of mitigation strategies requires overcoming the inherent challenges associated with traditional IAM-based approaches[30,31]. First, scenarios may exhibit model-determined biases, meaning that ensemble results can be influenced by

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https://doi.org/10.1038/s41558-025-02352-8

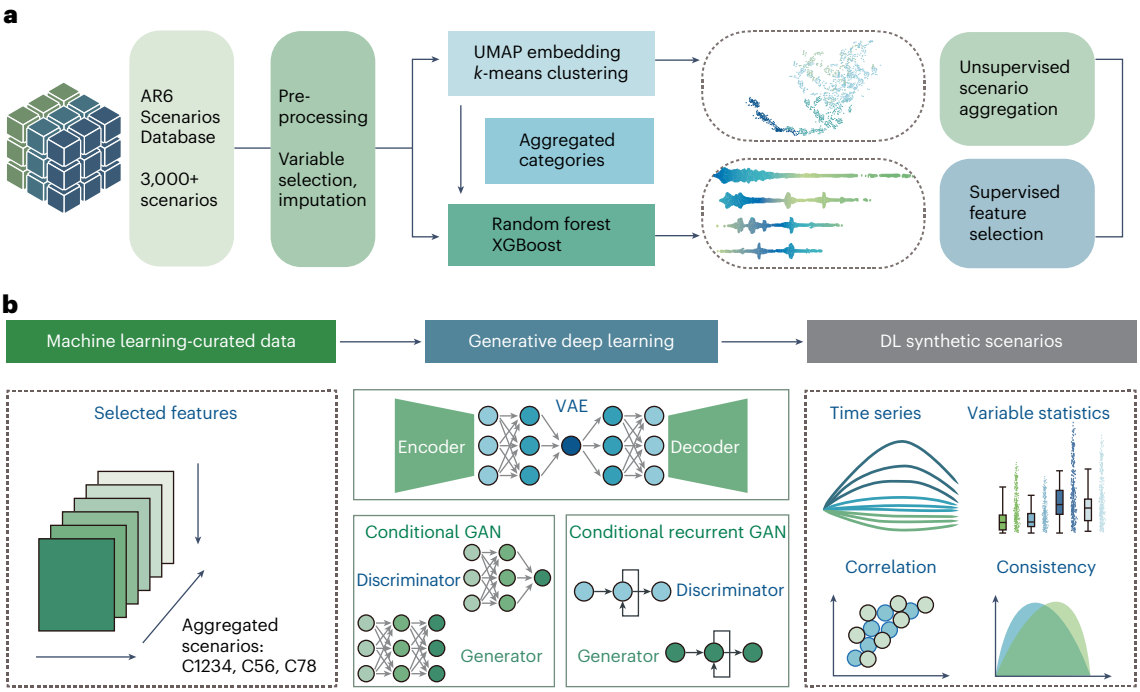


Fig. 1 | Overview of research framework. a, Dataset preparation: pre-processing steps involve variable selection and imputation to retain the 1,202 scenarios with IPCC climate categorization and 100 variables being reported by over 80% of scenarios for each 10-year interval. UMAP embedding and *k*-means clustering are used for unsupervised scenario aggregation. Random forest and XGBoost models are used for supervised feature selection. **b**, Generative DL models and scenario analysis: selected features are used to train generative DL models (VAE, CGAN and RCGAN) to generate synthetic scenarios. These scenarios are compared with the original AR6 scenarios for consistency and statistical validity. Detailed settings of each model are provided in Methods and Supplementary Text 3. Boxplots in **b** are illustrative only.

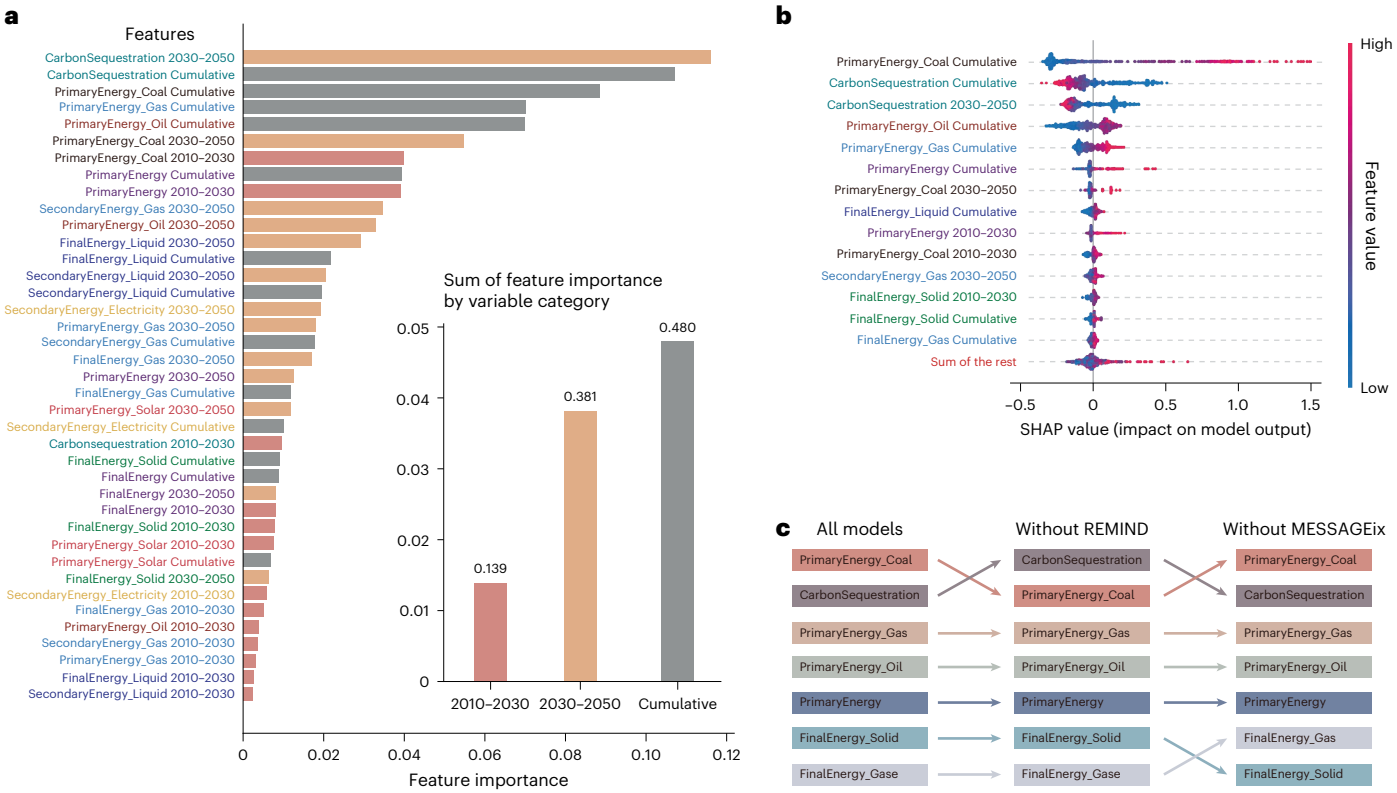


Fig. 2 | Identification of key determinants of mitigation scenarios. a, Key features ranked by the feature importance score of the random forest model. Texts are coloured by variables: coal, oil or liquids, gas, solid, solar, electricity, and carbon sequestration. Feature importance bars are coloured by the nature of variables: average change rate during 2010 to 2030, average change rate during 2030 to 2050, cumulative amount from 2010 to 2100. **b**, SHAP values of the top 14 key determinants. The directions and distributions indicate each variable's influence on outcomes from the C56 group as baseline. **c**, Top seven key features based on different sample sets used for feature ranking: scenarios including all models, excluding REMIND and excluding MESSAGEix. Arrows indicate changes in order.

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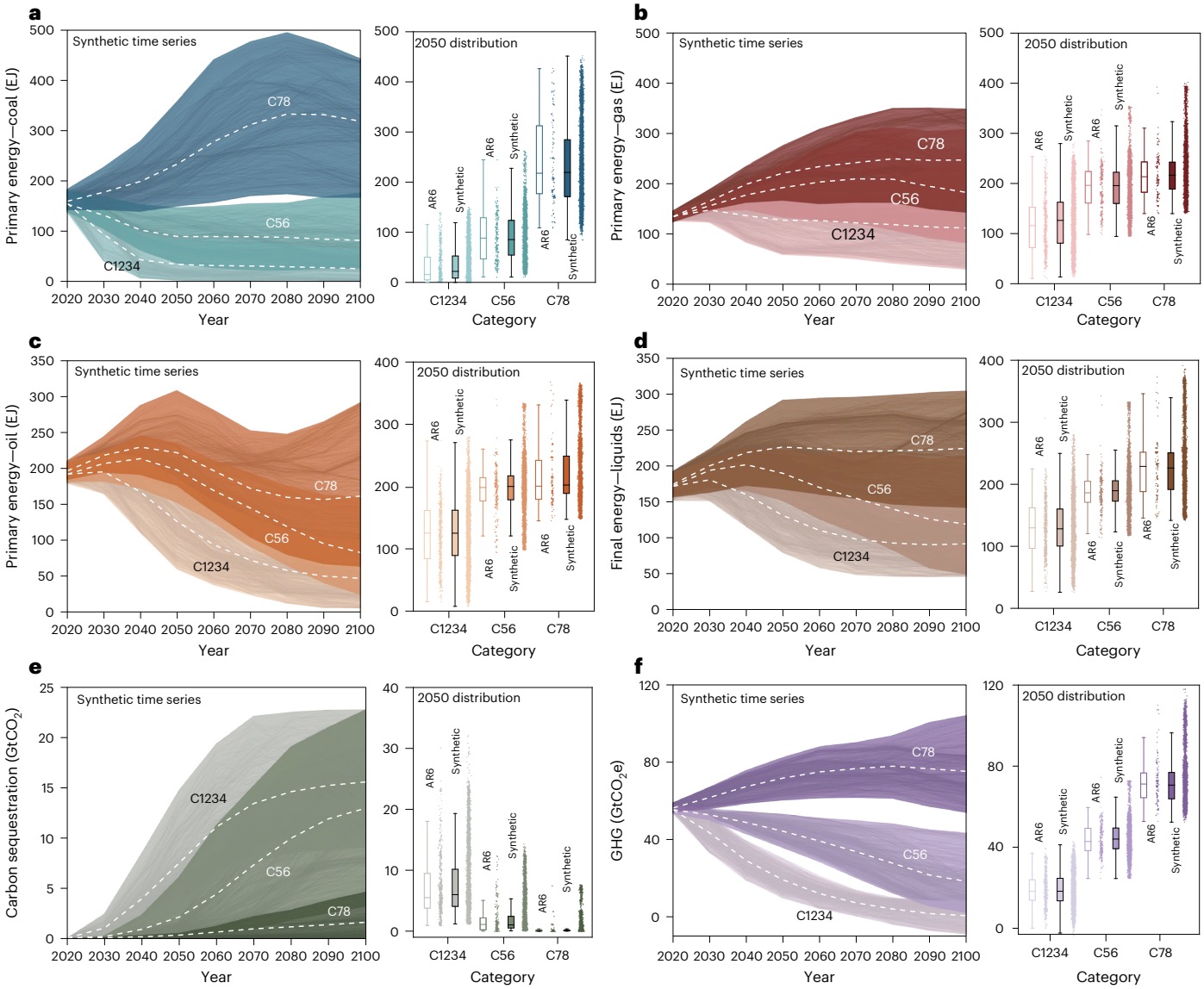


Fig. 3 | IPCC-consistent synthetic scenarios generated by VAE. a–f, Synthetic time series of primary coal consumption (**a**), primary gas consumption (**b**), primary oil consumption (**c**), final liquids consumption (**d**), carbon sequestration (**e**) and aggregated Kyoto gases (**f**) by VAE, compared to original raw data. Dashed lines represent the mean values of 10,000 synthetic scenarios for each of the C1234, C56 and C78 categories. The shaded area indicates the range of each category between the 10th and 90th percentiles. The grey line represents 1,000 sampled scenarios by categories. Detailed time series of these variables are shown in Supplementary Fig. 8. Boxplots summarize the distributions of variables (hollow/solid representing AR6/Synthetic, respectively). The central line within each box indicates the median, while the bottom and top edges correspond to the 25th and 75th percentiles (the interquartile range, IQR). The whiskers extend to the furthest data points within 1.5 times the IQR from the lower and upper quartiles. The same set of boxplots for 2100 is shown in Supplementary Fig. 9.

a model's idiosyncratic structures[32] and the non-random distribution of those voluntarily submitted scenarios[30]. While scenario databases undergo a rigorous vetting process to ensure quality and consistency, this process does not fully address biases introduced by selective submissions or modelling constraints. Therefore, it may lead to the underrepresentation of middle-of-the-pack scenarios between the well-known 'marker scenarios'[33] (such as climate stabilization scenarios between Representative Concentration Pathways (RCP) 4.5 and 2.6) as well as degrowth scenarios[34] that are less commonly modelled. Such limitations can restrict our understanding of the broad spectrum of policy-relevant pathways[35,36]. Second, existing practice for managing large-scale databases, ensuring quality control and categorizing scenarios has become increasingly complex and resource-intensive[37]. Addressing these issues requires the development of new computational approaches that complement existing methods by enhancing flexibility, scalability and inclusivity in scenario generation, while also ensuring consistency with established modelling practices.

Deep learning (DL) technologies have emerged as powerful tools for extracting hidden patterns and intricate interactions from high-dimensional data[38–41]. While discussions have illuminated DL's potential in a wide range of applications and outlined recommendations for its applications in climate change, the focus has largely been on overarching strategies and guiding principles, instead of directly generating mitigation scenarios. Moreover, unlike growing applications in physical systems such as weather forecasting[42], socioeconomic systems are shaped by human decisions, policy interventions and market dynamics, introducing unique challenges in capturing irrational behaviour and nonlinear interactions[43]. Here we present a generative DL model developed to generate synthetic scenarios spanning a range of future climate change mitigation ambitions, replicating key variable

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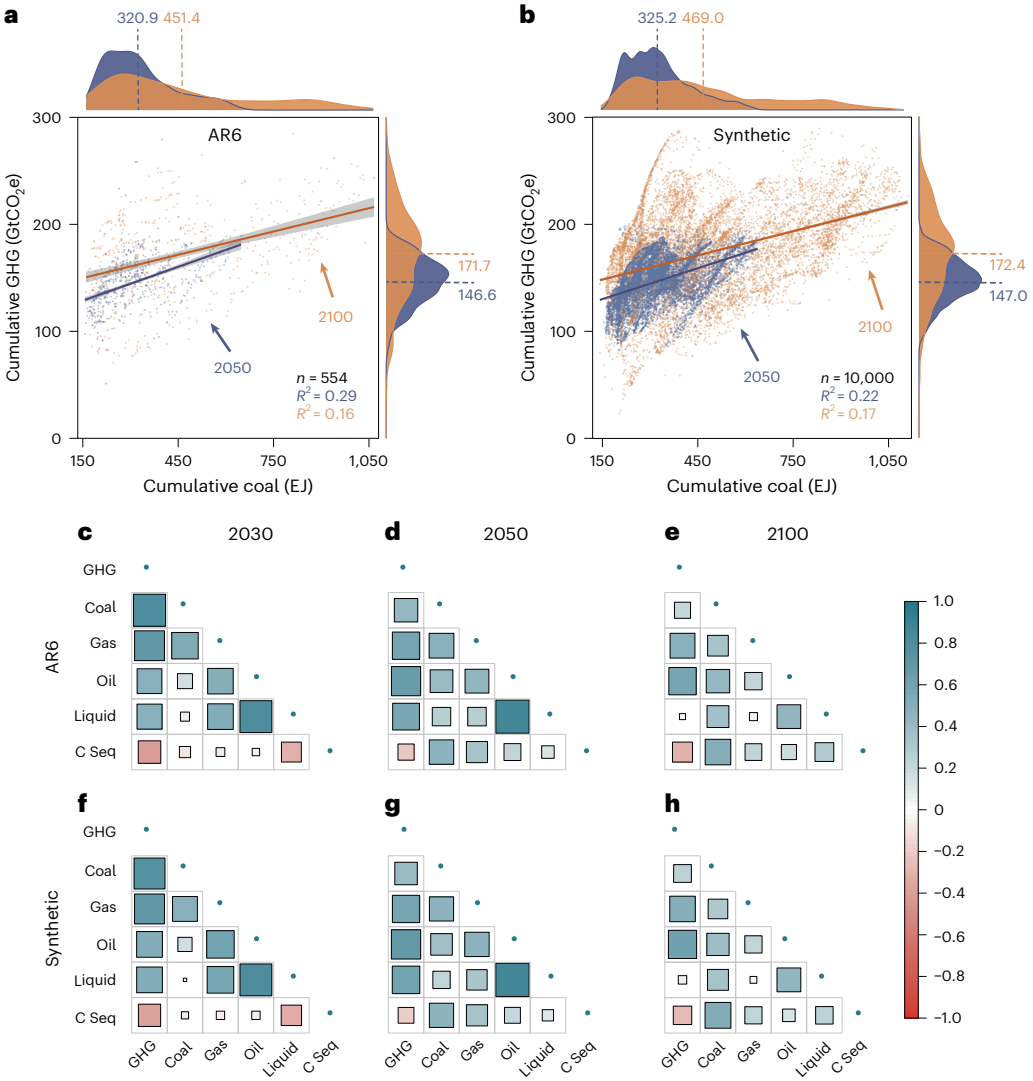


Fig. 4 | Correlations among key variables in AR6 scenarios and VAE-generated synthetic scenarios for the C1234 category. a,**b**, Scatterplots showing cumulative coal consumption (EJ) versus Kyoto gas emissions ($GtCO_2e$) over the periods 2020–2050 (dark blue) and 2020–2100 (orange) for AR6 scenarios (**a**) and VAE-generated synthetic scenarios (**b**). Dark blue and orange shadings denote the respective 2050 and 2100 timeframes. Error bands represents 95% confidence intervals. **c**–**e**, Correlation heatmaps of selected variables within the AR6 Scenarios Database for the years 2030 (**c**), 2050 (**d**) and 2100 (**e**). C Seq represents carbon sequestration. **f**–**h**, Correlation heatmaps of selected variables within the VAE-generated synthetic scenarios, illustrating similar patterns in variable interactions for the years 2030 (**f**), 2050 (**g**) and 2100 (**h**), compared to the AR6 Scenarios Database.

distributions and correlations while also demonstrating physical consistency in power sector variables. Hereinafter, we refer to mitigation scenarios generated using DL methods as synthetic scenarios, in contrast to IAM-based scenarios.

Our framework, detailed in Fig. 1 and the Methods section, aggregates IPCC AR6 mitigation scenarios[27] (C1–C8) into three categories (C1234, C56 and C78) through dimension reduction and clustering. C1–C8 are standard categories defined by the IPCC AR6, representing varying levels of climate outcomes, ranging from C1 (limit warming to 1.5 °C (>50%) with no or limited overshoot) to C8 (exceed warming of 4 °C ≥ 50%). We then use the feature importance score of the random forest classifier and the SHAP (SHapley Additive exPlanations) values of the XGBoost model to identify key variables and finally develop three generative DL models to produce synthetic scenarios, aligning with the original distribution of IPCC AR6 climate categories. Additionally, we evaluated the generated scenarios by assessing key variable distributions, correlations and internal consistency, providing insights into the strengths and limitations of the current framework while identifying opportunities for future improvements. Inspired by generative DL techniques in computer vision and natural language processing[44–47], the proof-of-concept approach aims to replicate IPCC AR6 mitigation scenarios, demonstrating how DL can complement traditional IAM methods by replicating key patterns and relationships.

Feature identification using interpretable machine learning

From the over 3,000 IAM-based scenarios in the AR6 Scenarios Database, we used the 1,202 scenarios with climate categorization for our training set. These scenarios have passed climate vetting process to ensure they meet AR6 reporting standards and contain sufficient emissions data to provide full-century climate outcomes. To enhance learning efficiency, we consolidated the original eight categories (C1–C8) into three groups, C1234, C56 and C78, aligning broadly with scenarios of Paris Agreement-aligned (low warming), medium warming and high warming scenarios. Detailed IPCC definitions of these categories and distribution across models can be found in Extended Data Table 1.

Using a random forest classifier, we identified key determinants for scenario grouping based on feature importance (Methods and Fig. 2a).

Fossil energy consumption, particularly coal, oil and gas, emerged as the highest-ranking features, highlighting the clear association between fossil energy use and climate change stabilization. Importantly, we found that the near-term carbon sequestration measures, such as afforestation, might be key players in climate change mitigation, showing the importance of early actions on coordinated carbon source and sink management. Overall, features related to changes between 2030 and 2050 are as important as cumulative amounts over the 2020–2100 period (see the inset in Fig. 2a). Next, we use SHAP values to provide explainable feature rankings (Fig. 2b), showing the directional effects on scenario categorization. Higher SHAP values (red) indicate a stronger association with variations in model output, confirming that higher cumulative coal consumption is associated with warmer temperature categories towards C78, while higher carbon sequestration correlates with more stringent climate targets towards C1234. These insights are also confirmed in actual IAM-based scenarios, such as their distributions in 2050 (Extended Data Fig. 1).

Our sensitivity analysis shows that top drivers of scenario categorization, such as coal consumption and carbon sequestration, remain consistent across IAMs, regardless of whether REMIND or MESSAGEix scenarios are excluded (Fig. 2c). Although 63% of the C1 scenarios were generated by REMIND and MESSAGEix, these robust feature rankings suggest that scenario categorization is driven more by variable patterns than by IAM-specific modelling idiosyncrasies (also see Extended Data Fig. 1).

DL-based synthetic mitigation scenarios

The training labels and features can also be used to generate synthetic scenarios consistent with IPCC categories. A variational autoencoder (VAE)[44] is a DL model that encodes data into a compressed representation and decodes it back to its original form, ideal for generating data points similar to the training set. Here we developed a DL framework based on VAE (labelled 'Synthetic' in Fig. 3) that closely matches the original AR6 scenarios ('AR6' in Fig. 3) in terms of mean and the spread, particularly for C1234 (deep decarbonization scenarios). While the high warming scenarios (C78) showed greater uncertainty due to varying assumptions across IAMs, in general synthetic scenarios have narrower distributions for all climate categories (boxplots in Fig. 3 and Extended Data Fig. 2), as VAE tends to converge to the centre of distribution, rather than replicating the outliers and extreme scenarios.

The synthetic scenarios also retained the correlations between key variables, crucial for maintaining internal consistency and physical relevance (Fig. 4), matching closely with those observed in the original AR6 scenarios. For C1234, the R^2 values between cumulative coal consumption and Kyoto gas emissions between 2020 and 2050 were 0.29 for AR6 and 0.22 for synthetic scenarios. These relatively low values are attributed to C1234 being low warming scenarios, where coal has been rapidly phased out, reducing its impact on Kyoto gas emissions. Furthermore, these correlations both diminished over longer periods (2020–2100) into 0.16 (AR6 scenarios) and 0.17 (synthetic scenarios), reflecting the phasing out of coal in C1234. Both AR6 and synthetic scenarios demonstrated stronger correlations between fossil consumption and Kyoto gas emissions in 2030 compared to 2100, confirming that near-term fossil fuel use remains a critical driver of GHG emissions, particularly in stringent mitigation scenarios. The similar patterns of multi-variable correlations in AR6 and synthetic scenarios suggest that our generative DL framework can to some degree capture complex, high-dimensional relationships without needing explicit physical constraint techniques.

Benchmarking with alternative DL methods, including conditional generative adversarial network (CGAN)[45,46] and recurrent conditional generative adversarial network (RCGAN)[47], revealed that VAE achieved higher category label transferring accuracy and better precision and recall (0.99 and 0.98, respectively) from synthetic data to real data, compared to RCGAN and CGAN (Methods, Table 1 and Extended Data Table 2, Experiment I). Precision measures the accuracy of positive predictions on climate categories made by the DL model, while recall assesses the model's ability to identify all relevant instances within a dataset. Here the term 'prediction' refers specifically to the label transfer task (Methods) and should not be conflated with model projections for mitigation scenarios, which have a distinct meaning in this context.

Table 1 | Summary of model accuracy comparing VAE, RCGAN and CGAN under label transferring tasks

Experiment I: training on synthetic scenarios		
	Testing set	**Accuracy**
VAE	AR6 scenarios	0.98
	Synthetic scenarios (self)	1.00
RCGAN	AR6 scenarios	0.75
	Synthetic scenarios (self)	1.00
CGAN	AR6 scenarios	0.69
	Synthetic scenarios (self)	1.00
Experiment II: training on AR6 scenarios		
	Testing set	**Accuracy**
VAE	AR6 scenarios (self)	1.00
	Synthetic scenarios	0.98
RCGAN	AR6 scenarios (self)	1.00
	Synthetic scenarios	0.99
CGAN	AR6 scenarios (self)	1.00
	Synthetic scenarios	1.00

In each experiment, a random forest classifier was trained and tested to assess the performance on label predictions on aggregated climate categories for C1234, C56 and C78. Accuracy represents the ratio of correctly predicted labels to the total number of samples.

Such transfer learning results provide further validation of the similarity between synthetic scenarios and the real IAM database. Different from VAE, CGAN involves a generator and a discriminator working in an adversarial manner to refine the quality of the synthetic data, while RCGAN builds on this by incorporating recurrent neural networks to capture sequential patterns, making it suitable for time series data. While CGAN and RCGAN performed better in generating concentrated samples in distribution (Extended Data Table 2, Experiment II), VAE proved more effective for our generative tasks to eliminate the overfitting effect and increase the breadth and coverage of scenarios, suggesting its suitability for data augmentation in well-structured scientific data.

Furthermore, as a demonstration, we assessed internal consistency in the power sector using a sum-check validation (Fig. 5). Sum-check is a common vetting metric in the AR6 Scenarios Database that ensures the total electricity generation matches the sum of its sub-fuel components. While we did not explicitly apply any constraints in our DL model, among the 30,000 synthetic scenarios, 258 exhibited lower sum-check errors than the 1.2% average error in the AR6 Scenarios Database, indicating that a subset of DL-generated scenarios meets or exceeds the internal consistency of IAM-based scenarios for this metric (Fig. 5a). Notably, while model training requires a similar time investment as running a single IAM scenario, once trained, DL models can generate 30,000 synthetic scenarios in seconds (Fig. 5b). Given such low computational costs, it is technically feasible to apply a large number of filtering criteria, provided the initial pool is sufficiently large. Additionally, we compared the power sector structure of synthetic scenarios with IAM-based scenarios, identifying the 'best-matched' IAM scenario for each climate category (C1234, C56, C78) (Fig. 5c–h). These results demonstrate that our prototype DL model is promising in replicating key sectoral trends and numeric values at an individual scenario level.

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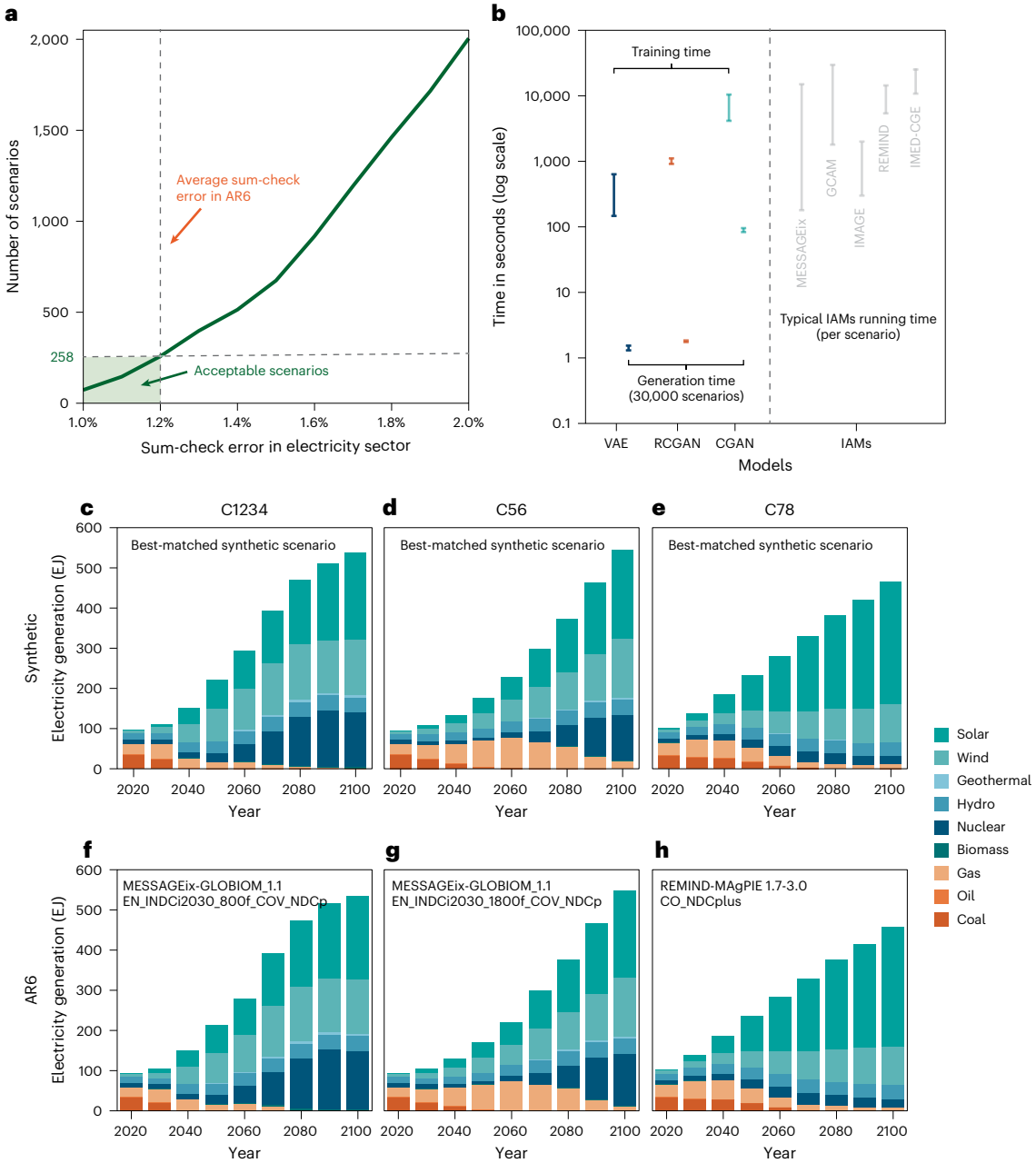


Fig. 5 | Internal consistency and computational performance of synthetic scenarios. a, Cumulative number of synthetic scenarios at various sum-check error levels in electricity sector. The sum-check error represents the relative difference between total electricity generation and the sum of individual sub-fuel contributions. **b**, Summary of model computational time. The left side shows the computational time for training the model and generating 30,000 synthetic scenarios using different DL models. Data ranges indicate different variable sets: the first includes the top six variables, while the second set extends this with an additional ten electricity sector variables. Training and generation times were recorded using an Intel(R) Core(TM) i9-13900HX processor (24 cores/32 threads, base frequency 2.20 GHz) and 16GB DDR5-5600MHz memory. The right side provides typical IAM running times, based on an expert survey, for comparison. **c**–**e**, Projection of electricity generation for the best-matched synthetic scenarios (VAE-generated scenarios with the lowest sum-check error) in C1234 (**c**), C56 (**d**) and C78 (**e**). **f**–**h**, Projection of electricity generation for the closest-matching AR6 scenarios in C1234 (**f**), C56 (**g**) and C78 (**h**), showing similar electricity generation structures over time.

Discussion

In climate change research, DL has been extensively applied to weather and climate modelling[48–51], power system planning[52,53] and energy demand forecasting[54–56]. Beyond conventional numerical data analysis, established language models such as BERT have been used to synthesize knowledge on climate change impacts, integrating textual data analysis into the broader scope of climate research[57]. More recently, there is a growing recognition of DL as a valuable tool in climate change mitigation research[58–60]. This analysis demonstrates that generative DL can effectively replicate key variables and relationships observed in IAM-based mitigation scenarios, capturing the interactions between energy system drivers and GHG emissions[61]. While previous DL applications focused on physical world processes, this work shifts the focus to the human–earth interactions underlying climate–economic scenarios traditionally analysed through IAMs.

Our results build on the existing literature on IAMs by demonstrating the feasibility of using DL techniques to replicate variable distributions, relationships and key sectoral results at an individual scenario

level. In this area, previous studies have applied machine learning techniques, such as random forest, to identify key drivers of scenario outcomes[62], similar to the analysis presented in Fig. 2. These insights into key drivers of mitigation pathways provide a critical foundation, which we leverage to develop generative AI tools as a potential avenue to overcome the data limitations identified in these earlier machine learning studies[62]. While our current approach primarily focuses on replication rather than full-scale complementarity, it provides a promising proof of concept for incorporating DL techniques into long-term climate change mitigation research. Given the low computational cost, such 'validation filtering' processes could be deployed to incrementally improve internal consistency across all sectors. Additionally, future research could extend this framework to model scenarios with region-specific information by incorporating regional socioeconomic drivers and sectoral details, addressing the underrepresentation of scenarios for missing regions within the scenario space and providing policymakers with context-specific mitigation pathways.

One key limitation of this study is that our generative DL framework replicates only a subset of key variables, far fewer than the hundreds of variables required for full-scale IAM scenarios. This restricts the ability of synthetic scenarios to fully capture the internal consistency of individual scenarios, as well as the multidimensional nature of policy implications, which often require detailed sectoral and technological representations. While we introduced sum-check validation as an initial step toward evaluating internal consistency, further improvements are needed to ensure broader physical and economic coherence across sectors. Additionally, the current framework does not yet incorporate socioeconomic feedback mechanisms, which are essential for capturing interactions between mitigation policies and socioeconomic drivers. Addressing these limitations will require several advancements: incorporating larger and more diverse training datasets to improve model generalization, exploring more complex DL architectures such as diffusion models to better capture high-dimensional relationships, and adopting physics-informed methods to physical consistency across variables.

Another notable limitation, often noted by IPCC authors and IAM modellers, is that scenario databases are considered 'ensembles of opportunities' rather than 'statistical samples'[31]. Previous studies have conducted a debiasing process by creating a sampled dataset with uniformly represented climate categories and models, instead of directly using all AR6 scenarios[28]. However, our DL models were trained on the existing AR6 Scenarios Database, which may retain or reinforce those biases, thus limiting the range of statistical distributions. Meanwhile, defining bias itself remains an open question—whether model-specific idiosyncrasies shape scenario distributions or certain patterns represent an inherent 'average' outcome remains a challenge that extends beyond statistical adjustments.

To improve scenario diversity and relevance to social needs, the IAM community has made substantial efforts, such as the updated Shared Socioeconomic Pathways (SSPs)[63], Scenario Model Intercomparison Project for CMIP7 (ScenarioMIP-CMIP7)[64] and the Scenario Compass Initiative (https://scenariocompass.org/), which improves the representation of key socioeconomic pathways. Building upon these structured, well-documented narratives that improve the representation of key socioeconomic drivers, generative DL models could further complement them by expanding scenario diversity while maintaining internal consistency. In particular, DL approaches such as model- or narrative-specific 'style transfer' could augment new SSPs by replicating marker scenarios with the unique structural signatures of different IAMs, broadening the representation of non-marker models and supporting a more comprehensive exploration of scenario uncertainty.

Looking forward, the integration of DL into climate change research holds substantial potential. Future work could develop more advanced data fusion and multimodal learning techniques that integrate diverse data sources, such as socioeconomic datasets and emission inventories to complement existing IAM scenarios. These advancements could also reduce the computational burden of expanding IAM scenario ensembles while improving scenario credibility and policy relevance. Moreover, explainable AI techniques should be incorporated to ensure that synthetic scenarios are interpretable, building trust with policymakers. Lessons from AI-driven climate modelling, where similar validation and transparency challenges have been addressed through model interpretability and uncertainty quantification techniques[65,66], can inform useful practices for socioeconomic scenario generation. This capability could be particularly beneficial for developing countries with limited resources, allowing them to integrate regional insights into their mitigation strategies.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41558-025-02352-8.

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Methods

Training set

In our study, we used the AR6 Scenarios Database supporting the IPCC AR6 Working Group III report. The AR6 Scenarios Database contains a total of 109 IAMs (including different versions of the same class of IAM) and 3,131 scenarios. For this work, we focused on a subset of 1,202 scenarios that passed the vetting process[67], ensuring these scenarios meet reporting standards and contain sufficient emissions data to provide comprehensive climate outcomes across categories C1 to C8. The number of scenarios in each category varies from 29 (C8) to 311 (C3) (Extended Data Table 1 and Supplementary Text 2). Category C1 represents limit warming to 1.5 °C (>50%) with no or limited overshoot; C2 represents return warming to 1.5 °C (>50%) after a high overshoot; C3 represents limit warming to 2 °C (>67%); C4 represents limit warming to 2 °C (>50%); C5 represents limit warming to 2.5 °C (>50%); C6 represents limit warming to 3 °C (>50%); C7 represents limit warming to 4 °C (>50%); and C8 represents exceed warming of 4 °C (≥ 50%). These scenarios are driven by socioeconomic exogenous inputs such as population and GDP and are modelled through endogenous variables such as energy consumption and use. These endogenous variables were identified as key drivers of scenario categorization and are the primary features considered in this analysis.

For feature selection and DL training, we assessed the completeness of variables reported by each scenario. Out of the 1,393 variables covering socioeconomics, energy, emissions and environment in the dataset, 100 variables met the criterion of being reported by over 80% of scenarios for each 10-year interval. These include variables of primary energy consumption, CO_2 emissions, electricity generation and population. In contrast, variables such as electricity capacity, investment, air pollutant emissions and water use have rates of over 50% missing. For the selected 100 variables, missing values were infilled using the sample mean for that specific year and climate category. For each variable, we keep their values from 2010 to 2100 with a 10-year time step. Additionally, we calculate the annual change rate during 2010–2030, 2030–2050 and 2050–2100 as additional temporal dimensions to capture more detailed policy-relevant dynamics in the near term.

Label identification

We consolidated the eight original IPCC categories into three aggregated groups (C1234, C56 and C78) to manage small sample sizes in several categories. First, we calculate the dynamic time warping (DTW) distance[68] between any pair of trajectories to quantify the similarities between scenarios. Second, we use uniform manifold approximation and projection (UMAP)[69], with n_neighbors set to 15, to reduce the dimensionality of these DTW matrices. This process projected each trajectory onto a two-dimensional plane, where closer data points indicate stronger similarities or relationships between the scenarios. Finally, we implemented *k*-means clustering on the UMAP-reduced space, iteratively reducing n_clusters from the original eight IPCC categories to three, which in general corresponds to a direct aggregation of C1234, C56 and C78, with 66.3%, 87.9% and 96.7% of scenarios mapped in the same category (Supplementary Fig. 2 and Supplementary Text 2). For simplicity, we then directly use C1234, C56 and C78 aggregation as our labels. We also performed our analysis using the original eight IPCC categories, and model performance is decreased due to small sample sizes in each category (Supplementary Text 1).

Identification of key features

To determine the importance of model features, we constructed random forest models for the aggregated labels (Fig. 2). A random forest model is an ensemble learning method that constructs multiple decision trees to identify important variables based on their contribution to the variance explained in the data. We randomly split the data into an 80% training set and a 20% test set (using the train_test_split function from the sklearn library with parameters test_size = 0.2, random_state = 3). For each model, we compressed time series of feature variables into cumulative quantities and temporal changes (2010–2030, 2030–2050 and 2050–2100) and incorporated additional model-specific features. By using the feature importance attribute of the random forest model, we identified key variables and ranked them, as shown in Fig. 2a. One limitation we noted is the potential for collinearity among variables, such as when multiple variables are present for the same feature, which may result in them all being ranked highly due to their interrelated effects.

To further check the consistency of identified feature variables, we additionally used SHAP[70] to quantify feature contribution to the variations in aggregated climate categories (Fig. 2b). SHAP values are a method derived from game theory to explain the output of machine learning models, providing insights into the contribution of each feature to the aggregated climate categories. We constructed an XGBoost model[71] (with parameters objective = 'reg:squarederror', max_depth = 6, reg_alpha = 0.2, random_state = 5, n_estimators = 100) with Kyoto gas emissions as the response variable (normalized into values between 0 and 1) and used the SHAP method to interpret the model. The top-ranking variables identified through SHAP were found to be consistent with the variables selected by the random forest model (Supplementary Text 2). We also performed the same random forest-based feature ranking excluding scenarios from MESSAGEix or REMIND, and the top features remained largely consistent (Fig. 2c).

Generative DL models for synthetic scenarios

We employed three representative generative models using the TSGM library[72]: VAE, CGAN and RCGAN. For demonstration, all three methods are tasked to produce 30,000 synthetic scenarios, with 10,000 for each climate category. Synthetic scenarios included temporal trajectories of key determinants of climate categories (Kyoto gases, primary energy consumptions of coal, gas and oil, final energy consumption of refined liquids, and carbon sequestrations), as well as all fuel-level variables in the electricity sector for the evaluation of consistency. It is important to note that, while these synthetic scenarios replicate key variables observed in IAM-based pathways, they do not reconstruct full IAM outputs, which include broader and more complex cross-sector interdependencies. Features were normalized using tsgm.utils.TSFeatureWiseScaler((0,1)) to ensure consistent scaling across features (Supplementary Text 3). DL model training and generations were conducted using an Intel(R) Core(TM) i9-13900HX processor (24 cores/32 threads, base frequency 2.20 GHz) and 16GB DDR5-5600MHz memory. Model training and computation times are shown in Fig. 5b.

The VAE takes advantage of the idea of the autoencoder model to map the raw data into a low-dimensional latent space that preserves the essential information and structure of the original dataset. In the autoencoder, there is an encoder and a decoder structure to self-supervise the learning of latent space. The neural network-based encoder compresses the input data into a latent representation, and another neural network decoder maps the latent representation back to the reconstructed data. Different from the autoencoder, which is mainly targeted for dimensionality reduction, the VAE achieves the data generation capability by making the encoder no longer directly output the latent representation, but instead output the parameters of the latent variable distribution (generally assuming multivariate normal distribution for the ease of sampling), usually the mean and variance of the distribution, that is $\mu, \log \sigma^2 = f_{\text{enc}}(x)$. To generate a new observation, another latent variable is sampled in the latent space via the reparameterization trick $z = \mu + \sigma \odot \epsilon$, $\epsilon \sim \mathcal{N}(0, I)$, where ϵ is sampled from a standard normal distribution and $\odot$ denotes element-wise multiplication. The latent data are then sent to the decoder, which maps the latent data into the original data space based on the sampling results, that is $\hat{x} = f_{\text{dec}}(z)$. We trained VAE models separately on three aggregated categories (C1234, C56, C78) (detailed model settings are provided in Extended Data Table 3) and conducted comparisons based

on boxplots for different variables between synthetic data and real data (Fig. 3).

Additionally, we trained a CGAN model with parameters provided in Supplementary Text 3. CGAN optimizes the generator $G(z|c)$ and discriminator $D(x|c)$ through adversarial training, gradually improving the quality of the synthetic data. Here x represents the real data sample and z is the random noise input, which acts as the seed for generating synthetic data, while c represents the conditioning variable such as categories. Specifically, during training process, the generator, parametrized by neural network with learnable parameters, continuously improves the quality of the synthetic data, causing the discriminator to misclassify the synthetic data as real data, and the neural network-based discriminator continuously improves its discrimination ability to better distinguish between real data and synthetic data. The process could be described by following mathematical formula:

$$\min_G \max_D E_{x\sim p_{\text{data}}(x)}\left[\log D\left(x|c\right)\right] + E_{z\sim p_z(z)}\left[\log\left(1 - D\left(G\left(z|c\right)|c\right)\right)\right] \quad (1)$$

Through such an adversarial training process, the synthetic scenarios become increasingly realistic and indistinguishable from actual observations. In addition, the 'conditional' design of the CGAN model allows more flexible guidance of the data generation process by providing specific aggregated categories.

Finally, we trained the RCGAN model to better cope with the temporal structure of data. Compared to CGAN, the classical neural network design of the generator $G(z|c)$ and discriminator $D(x|c)$ in the RCGAN model was specified by RNN (recurrent neural network), which was more suitable for generating sequence data and capturing temporal correlations. The RNN structure functions by remembering and aggregating variable information from previous time steps when modelling the next one, especially useful when involving the cumulation effects. The RNN process can be described with the following formula:

$$\mathbf{h}_t = \mathbf{f}\left(W_h\mathbf{h}_{t-1} + W_x\mathbf{x}_t + \mathbf{b}\right) \quad (2)$$

where $\mathbf{h}_t$ is a vector representing the hidden state at the current time step t; it serves as a summary of all cumulated information learned by the model from the time series up to that point. The term $\mathbf{h}_{t-1}$ is the hidden state from the immediately preceding time step $t-1$ effectively denoting the model's memory of past variables. The variable $\mathbf{x}_t$ represents the input variables at the current time step t. The matrices W_h and W_x are part of the model's parameters, which are learnable weights used to quantify influence: specifically, W_h is the weight matrix associated with the previous hidden state $\mathbf{h}_{t-1}$, modulating the impact of past states, while W_x is the weight matrix associated with the current input variables $\mathbf{x}_t$, determining the influence of current inputs. Finally, $\mathbf{b}$ is a bias term that introduces an offset, allowing for greater flexibility in the model's predictions, and $\mathbf{f}$ is the nonlinear activation function operating elementwise, which ensures the model can capture complex relationships in the sequence. By combining the advantages of conditional generation based on aggregated labels with the improved ability to model temporal dynamics, the RCGAN model provides the capability to realistically generate scenarios over time.

Benchmarking generative models on label transfer task

To benchmark these generative models, we trained a random forest model on both the DL-generated dataset and the original AR6 Scenarios Database and compared the label transfer performance between them (Table 1). Here we apply the label transfer task, which originated from the idea of domain adaption in transfer learning[73], to train a classification model on one dataset and then evaluate its performance on another by comparing the true labels with pseudo labels predicted by the classifier. The method can assess how well the features and patterns learned from the first dataset generalize to the second. If the classifier trained on real data achieved high accuracy on the generated data, this indicates that the generated data has captured the most important features of the real dataset. Conversely, if the classifier trained on synthetic data also performed well on the real dataset, it further demonstrates the breadth and coverage of the synthetic data in real IAM-based scenarios.

Here we found that the classification model trained on the real dataset demonstrated high accuracy when predicting the generated dataset regardless of generation methods (Extended Data Table 2, Experiment II), indicating effective capture of interactions between input variables and carbon emissions. However, when the model trained on the synthetic dataset predicted the real dataset, accuracy decreased substantially for all generative methods (Extended Data Table 2, Experiment I), and the data tended to concentrate in high-density regions, indicating a potential overfitting issue in the model trained with synthetic data. Notably, VAE achieved the best overall performance when transferring labels from synthetic data toward the real dataset, indicating its robustness for data generation tasks.

Internal consistency

For the demonstration of internal consistency, in addition to selecting variables based on feature importance from the random forest model, we also incorporated fuel-level variables in the electricity sector (SecondaryEnergy|Electricity), including SecondaryEnergy|Electricity, SecondaryEnergy|Electricity|Biomass, SecondaryEnergy|Electricity|Coal, SecondaryEnergy|Electricity|Gas, SecondaryEnergy|Electricity|Geothermal, SecondaryEnergy|Electricity|Hydro, SecondaryEnergy|Electricity|Nuclear, SecondaryEnergy|Electricity|Oil, SecondaryEnergy|Electricity|Solar and SecondaryEnergy|Electricity|Wind, into our generative model. This ensures that the scenarios generated by the models (VAE, CGAN, RCGAN) are representative of the entire electricity system. We used the VAE model to generate 10,000 samples per category (C1234, C56, C78), resulting in a total of 30,000 samples. To evaluate consistency, we performed a sum-check in the electricity sector, a vetting criterion commonly used in scenario databases. The sum-check error represents the relative difference between the total electricity generation and the sum of all fuel-levels generation, calculated using the following formula:

$$\text{Error} = \frac{\left|\text{Gen}_{\text{total}} - \sum_i \text{Gen}_i\right|}{\text{Gen}_{\text{total}}} \quad (3)$$

where $\text{Gen}_{\text{total}}$ is the total electricity generation from the synthetic scenarios, Gen_i is the electricity generation from fuel type i. The sum-check error serves as a demonstration of consistency in the electricity generation structure across different fuel sources. Finally, we validated the scenarios by comparing the time series of the best-matched scenarios—those with the minimum sum-check error among VAE-generated scenarios—with the closest counterparts from the AR6 Scenarios Database (Fig. 5c–h).

Data availability

The AR6 Scenarios Database (ref. 27) is publicly available at https://iiasa.ac.at/models-tools-data/ar6-scenario-explorer-and-database. VAE-generated scenarios (n = 30,000) are available via Zenodo at https://doi.org/10.5281/zenodo.15240553 (ref. 74). Source data are provided with this paper.

Code availability

Source code to reproduce this analysis is available via Zenodo at https://doi.org/10.5281/zenodo.15280653 (ref. 75).

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Acknowledgements

This work was supported by the National Key R&D Program of China (grant no. 2024YFF1307000), the National Natural Science Foundation of China (grant no. 72474002), Emerging Engineering Interdisciplinary Young Scholars Project, Peking University, and the Fundamental Research Funds for the Central Universities. H.M. was supported by the National Research Foundation of Korea (grant no. RS-2024-00467678). DL model training was supported by the High-Performance Computing Platform of Peking University. E.B. received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement no. 821471 (ENGAGE).

Author contributions

Y.O. and P.Z. jointly designed the study. P.Z. supervised the DL techniques. Y.O. supervised the IAM data analysis. E.B. contributed to modelling data compilation. H.M. contributed to data interpretation. P.L. processed the data and developed the DL models. P.L., R.Z. and Y.O. wrote the paper with input from all co-authors.

Competing interests

The authors declare no competing interests.

Additional information

Extended data is available for this paper at https://doi.org/10.1038/s41558-025-02352-8.

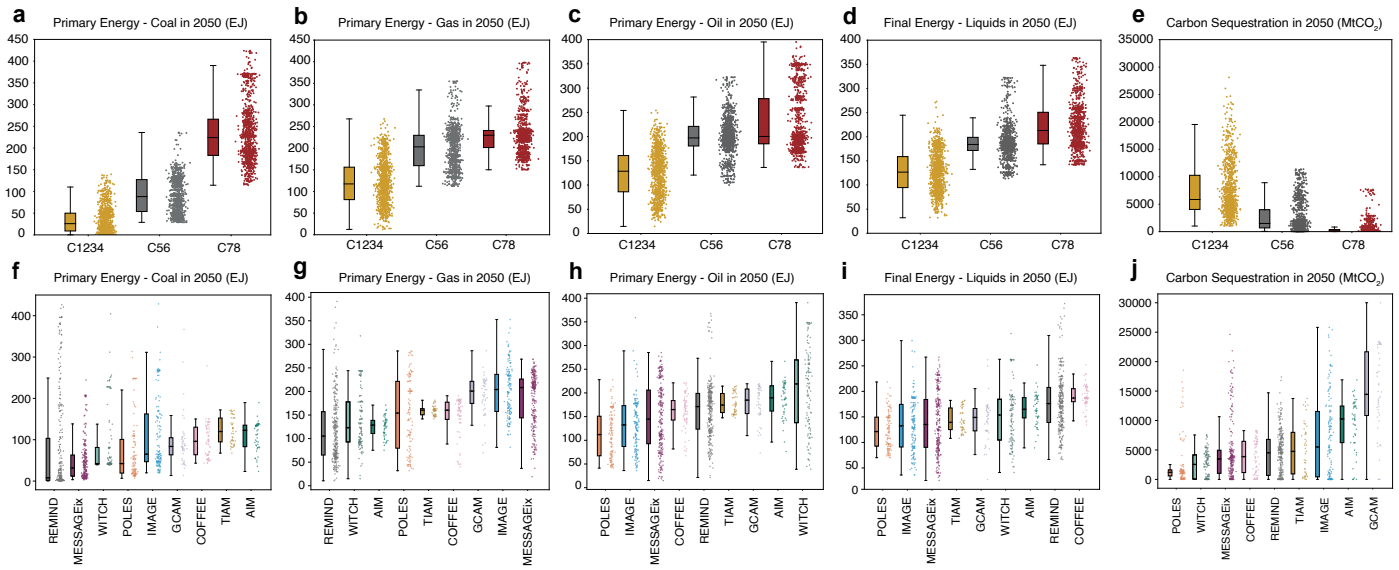
Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41558-025-02352-8.

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Peer review information *Nature Climate Change* thanks Alaa Al Khourdajie, Nikola Milojevic-Dupont and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

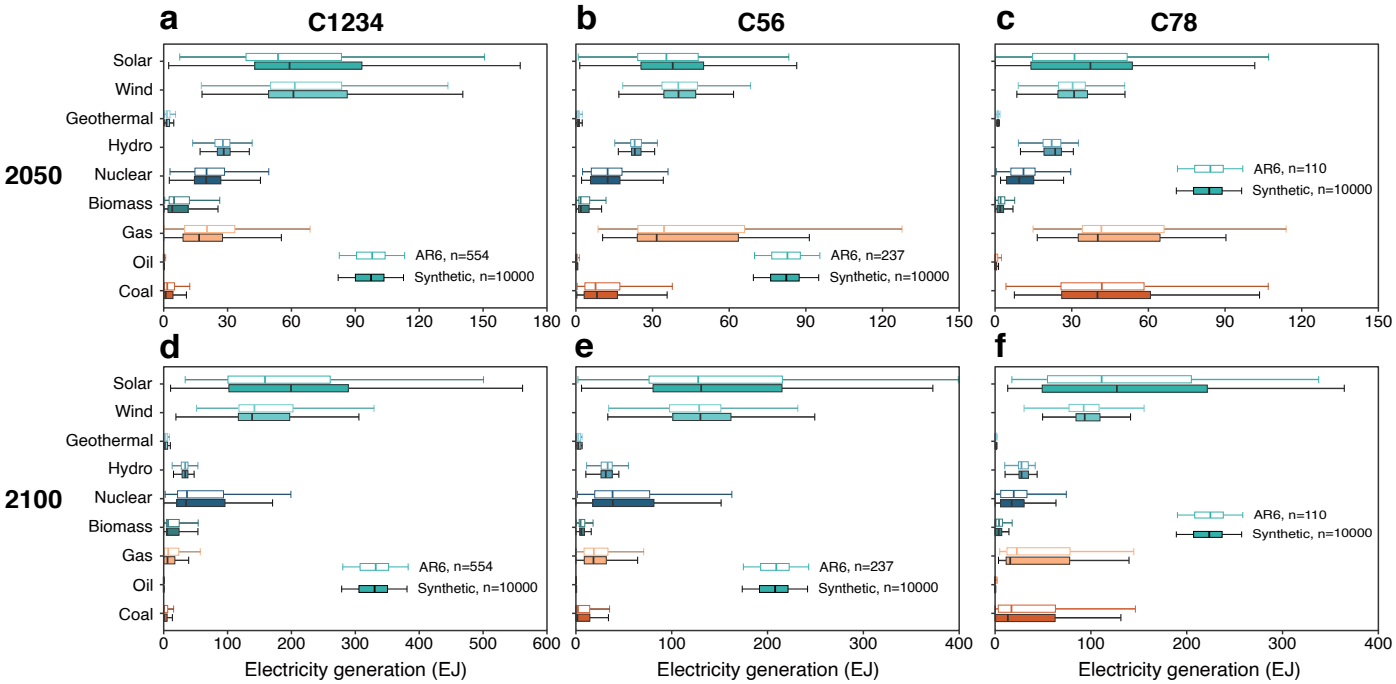
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https://doi.org/10.1038/s41558-025-02352-8



Extended Data Fig. 1 | Distribution of selected key features in 2050 in the AR6 Scenarios Database. a-e, Boxplots of primary coal consumption (EJ), primary gas consumption (EJ), primary oil consumption (EJ), final liquids consumption (EJ), and carbon sequestration ($MtCO_2$) by aggregated scenario categories of C1234, C56, and C78. **f-j**, Boxplots of the same variables grouped by the type of IAMs. The central line within each box indicates the median, while the bottom and top edges correspond to the 25th and 75th percentiles (the interquartile range, IQR). The whiskers extend to the furthest data points within 1.5 times the IQR from the lower and upper quartiles.

https://doi.org/10.1038/s41558-025-02352-8



Extended Data Fig. 2 | Distribution of electricity sector variables in 2050 and 2100 in the AR6 Scenarios Database (blank boxes) and synthetic scenarios (solid boxes). a-c, Boxplots of electricity generation by fuels by aggregated scenario categories of C1234, C56, and C78 in 2050. **d-f**, Boxplots of the same variables in 2100. The central line within each box indicates the median, while the bottom and top edges correspond to the 25th and 75th percentiles (the interquartile range, IQR). The whiskers extend to the furthest data points within 1.5 times the IQR from the lower and upper quartiles.

https://doi.org/10.1038/s41558-025-02352-8

Extended Data Table 1 | Summary of scenarios passed climate vetting by climate category and submission models

Model	IPCC climate category								
	C1	C2	C3	C4	C5	C6	C7	C8	Total
REMIND	41	44	84	16	34	19	48	11	297
MESSAGEix	20	43	59	39	57	20	28	0	266
IMAGE	7	9	34	18	22	16	34	2	142
WITCH	9	2	31	14	24	9	12	14	115
POLES	4	10	26	24	20	11	19	0	114
COFFEE	1	4	14	15	21	9	1	0	65
AIM	4	3	17	8	13	4	6	0	55
GCAM	6	6	13	9	6	1	6	1	48
TIAM	0	0	20	6	10	4	5	0	45
GEM	2	10	12	6	5	3	3	0	41
EPP	0	0	1	3	0	1	2	0	7
CRO	3	2	0	0	0	0	0	1	6
MER	0	0	0	1	0	0	0	0	1
Total	97	133	311	159	212	97	164	29	**1202**

C1 category means limit warming to 1.5 °C (>50%) with no or limited overshoot, C2 category means return warming to 1.5 °C (>50%) after a high overshoot, C3 category means limit warming to 2 °C (>67%), C4 category means limit warming to 2oC (>50%), C5 category means limit warming to 2.5 °C (>50%), C6 category means limit warming to 3 °C (>50%), C7 category means limit warming to 4 °C (>50%), C8 category means exceed warming of 4 °C (≥50%).

https://doi.org/10.1038/s41558-025-02352-8

Extended Data Table 2 | Comparison of label transfer performance of different generative deep learning methods in two experimental conditions

Experiment I: training on synthetic scenarios and testing on AR6 scenarios					
	Category	**Precision**	**Recall**	**F1-score**	**Testing sample**
VAE	C1234	0.99	0.99	0.99	554
	C56	0.97	0.97	0.97	237
	C78	1.00	0.97	0.99	110
	macro avg	0.99	0.98	0.98	901
	weighted avg	0.98	0.98	0.98	901
RCGAN	C1234	0.85	0.85	0.85	554
	C56	0.53	0.59	0.56	237
	C78	0.86	0.61	0.71	110
	macro avg	0.75	0.69	0.71	901
	weighted avg	0.76	0.75	0.76	901
CGAN	C1234	0.93	0.70	0.80	554
	C56	0.75	0.53	0.62	237
	C78	0.34	0.97	0.50	110
	macro avg	0.67	0.73	0.64	901
	weighted avg	0.81	0.69	0.72	901
Experiment II: training on AR6 scenarios, testing on synthetic scenarios					
	Category	**Precision**	**Recall**	**F1-score**	**Testing sample**
VAE	C1234	0.98	0.99	0.98	1000
	C56	0.95	0.98	0.96	1000
	C78	1.00	0.96	0.98	1000
	macro avg	0.98	0.98	0.98	3000
	weighted avg	0.98	0.98	0.98	3000
RCGAN	C1234	1.00	1.00	1.00	1000
	C56	0.98	0.99	0.99	1000
	C78	0.99	0.98	0.99	1000
	macro avg	0.99	0.99	0.99	3000
	weighted avg	0.99	0.99	0.99	3000
CGAN	C1234	1.00	1.00	1.00	1000
	C56	1.00	0.99	1.00	1000
	C78	1.00	1.00	1.00	1000
	macro avg	1.00	1.00	1.00	3000
	weighted avg	1.00	1.00	1.00	3000

Experiment I from synthetic to real data (synthetic scenarios for training, AR6 scenarios for testing) and Experiment II from real to synthetic data (AR6 scenarios for training, synthetic scenarios for testing). In each experiment, a Random Forest classifier was trained and tested to assess the performance on label predictions for C1234, C56, and C78. Precision measures the accuracy of positive predictions made by the model, while Recall assesses the model's ability to identify all relevant instances within a dataset. 'Macro avg' represents the average of three categories for each metric, and 'weighted avg' represents the average of three categories for each metric weighted by sample size.

https://doi.org/10.1038/s41558-025-02352-8

Extended Data Table 3 | Detailed architecture and parameters in the VAE model

Model	Optimizer	Decayed learning rate	Parameter	Epochs	Batch size
VAE_C1234	Adam	Yes	initial_learning_rate=1e-3, decay_steps=500, decay_rate=0.98	3,000	50
VAE_C56	Adam	Yes	initial_learning_rate=1e-3, decay_steps=200, decay_rate=0.98	3,000	30
VAE_C78	Adam	Yes	initial_learning_rate=1e-3, decay_steps=200, decay_rate=0.98	3,000	30
RCGAN_ C1234	Adam	Yes	initial_learning_rate=1e-3, decay_steps=200, decay_rate=0.95	1,800	100
RCGAN_ C56	Adam	Yes	initial_learning_rate=1e-3, decay_steps=200, decay_rate=0.95	1,800	100
RCGAN_ C78	Adam	Yes	initial_learning_rate=1e-3, decay_steps=200, decay_rate=0.95	1,800	100
CGAN_ C1234	Adam	Yes	initial_learning_rate=1e-4, decay_steps=200, decay_rate=0.95	1,500	100
CGAN_ C56	Adam	Yes	initial_learning_rate=1e-4, decay_steps=200, decay_rate=0.95	1,500	100
CGAN_ C78	Adam	Yes	initial_learning_rate=1e-4, decay_steps=200, decay_rate=0.95	1,500	100

Step-by-step model setup is documented in Supplementary Text 3.